

Homework 6: Gram-Schmidt Process

BCS A, E, F, G

Problem. *Legendre Polynomials.*

Let the vector space P_2 have the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx .$$

Apply the Gram–Schmidt process to transform the standard basis $\{1, x, x^2\}$ for P_2 into an orthogonal basis $\{\phi_1(x), \phi_2(x), \phi_3(x)\}$.

Remark. The resulting polynomials are known as the first three *Legendre polynomials*, play an important role in a variety of applications. The scaling does not affect the orthogonality.

Submission Date: October 11th, 2024 (11:59PM)

Good Luck